

task4-Step-by-Step Guide to Create linux-practice.md

You'll perform a few Linux commands, observe the output, and write everything into a markdown document.

1. Process Checks Section

Add this heading:

□ `## Process Checks`

□ Now run these commands in terminal.

Command 1 — View Running Processes

□ `ps aux | head`

□ This shows currently running processes.

```
ubuntu@ip-172-31-32-12:~$ ps aux | head
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root             1  0.2  1.7  25196 16004 ?        Ss   06:00   0:02 /sbin/init
root             2  0.0  0.0      0      0 ?        S    06:00   0:00 [kthreadd]
root             3  0.0  0.0      0      0 ?        S    06:00   0:00 [pool_workqueue_release]
root             4  0.0  0.0      0      0 ?        I<   06:00   0:00 [kworker/R-rcu_gp]
root             5  0.0  0.0      0      0 ?        I<   06:00   0:00 [kworker/R-sync_wq]
root             6  0.0  0.0      0      0 ?        I<   06:00   0:00 [kworker/R-kvfree_rcu_reclaim]
root             7  0.0  0.0      0      0 ?        I<   06:00   0:00 [kworker/R-slub_flushwq]
root             8  0.0  0.0      0      0 ?        I<   06:00   0:00 [kworker/R-netns]
root            10  0.0  0.0      0      0 ?        I<   06:00   0:00 [kworker/0:0H-kblockd]
ubuntu@ip-172-31-32-12:~$
```

Command 2 — Search Specific Process

□ `pgrep systemd`

OR

top

If using top, press q to quit.

Record output similarly.

```
top - 06:21:17 up 21 min, 1 user, load average: 0.00, 0.00, 0.00
Tasks: 114 total, 1 running, 113 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.2 us, 0.0 sy, 0.0 ni, 99.8 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 908.7 total, 380.9 free, 304.7 used, 332.6 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 604.0 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
34	root	20	0	0	0	0	I	0.3	0.0	0:00.07	kworker/u8:1-events_power_efficient
1	root	20	0	25196	16004	10916	S	0.0	1.7	0:02.46	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/0:0H-kblockd
12	root	20	0	0	0	0	I	0.0	0.0	0:00.16	kworker/u8:0-events_power_efficient
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.00	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:00.04	rcu_sched
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
23	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
24	root	rt	0	0	0	0	S	0.0	0.0	0:00.00	migration/1
25	root	20	0	0	0	0	S	0.0	0.0	0:00.00	ksoftirqd/1
27	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/1:0H-kblockd

4. Service Checks Section

Add:

Service Checks

Choose one service available on your system.

Common examples:

- ssh
- cron
- docker
- nginx

Command 3 — Check Service Status

Example using SSH:

```
systemctl status ssh
```

Copy important lines into markdown.

```
ubuntu@ip-172-31-32-12:~$ systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Drop-In: /usr/lib/systemd/system/ssh.service.d
            └─ec2-instance-connect.conf
   Active: active (running) since Wed 2026-05-20 06:00:14 UTC; 22min ago
  Invocation: df37663904a4414cad8647b621aad15e
 TriggeredBy: ● ssh.socket
    Docs: man:ssh(8).
          man:ssh_config(5).
   Main PID: 1015 (sshd)
     Tasks: 1 (limit: 627)
    Memory: 7.5M (peak: 14.8M)
       CPU: 1.076s
    CGroup: /system.slice/ssh.service
            └─1015 "sshd: /usr/sbin/sshd -D -o AuthorizedKeysCommand /usr/share/ec2-instance
```

Command 4 — List Running Services

```
systemctl list-units --type=service --state=running
```

```
ubuntu@ip-172-31-32-12:~$ systemctl list-units --type=service --state=running
UNIT                                LOAD    ACTIVE SUB    DESCRIPTION
acpid.service                       loaded active running ACPI event daemon
chrony.service                      loaded active running chrony, an NTP client/server
cron.service                        loaded active running Regular background program processing daemon
dbus.service                        loaded active running D-Bus System Message Bus
getty@tty1.service                  loaded active running Getty on tty1
irqbalance.service                 loaded active running irqbalance daemon
ModemManager.service                loaded active running Modem Manager
multipathd.service                  loaded active running Device-Mapper Multipath Device Controller
networkd-dispatcher.service          loaded active running Dispatcher daemon for systemd-networkd
polkit.service                      loaded active running Authorization Manager
rsyslog.service                     loaded active running System Logging Service
serial-getty@ttyS0.service           loaded active running Serial Getty on ttyS0
snap.amazon-ssm-agent.amazon-ssm-agent.service loaded active running Service for snap application amazon-ssm-agent.amazon-s
snapd.service                       loaded active running Snap Daemon
ssh.service                         loaded active running OpenBSD Secure Shell server
systemd-journald.service             loaded active running Journal Service
systemd-logind.service               loaded active running User Login Management
systemd-networkd.service             loaded active running Network Management
systemd-resolved.service             loaded active running Network Name Resolution
systemd-udev.service                loaded active running Rule-based Manager for Device Events and Files
udisks2.service                     loaded active running Disk Manager
unattended-upgrades.service          loaded active running Unattended Upgrades Shutdown
user@1000.service                   loaded active running User Manager for UID 1000

Legend: LOAD    → Reflects whether the unit definition was properly loaded.
         ACTIVE → The high-level unit activation state, i.e. generalization of SUB.
         SUB    → The low-level unit activation state, values depend on unit type.

23 loaded units listed.
```

5. Log Checks Section

Add:

Log Checks

Command 5 — View Logs for Service

Example:

journalctl -u ssh --no-pager | tail -n 20

```
23 loaded units listed.
ubuntu@ip-172-31-32-12:~$ journalctl -u ssh --no-pager | tail -n 20
May 20 06:00:14 ip-172-31-32-12 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
May 20 06:00:14 ip-172-31-32-12 sshd[1015]: Server listening on 0.0.0.0 port 22.
May 20 06:00:14 ip-172-31-32-12 sshd[1015]: Server listening on :: port 22.
May 20 06:00:14 ip-172-31-32-12 systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
May 20 06:01:30 ip-172-31-32-12 ec2-instance-connect[1254]: Querying EC2 Instance Connect keys for matching finger
May 20 06:01:30 ip-172-31-32-12 ec2-instance-connect[1286]: Providing ssh key from EC2 Instance Connect with finger
May 20 06:01:30 ip-172-31-32-12 ec2-instance-connect[1403]: Querying EC2 Instance Connect keys for matching finger
May 20 06:01:31 ip-172-31-32-12 sshd-session[1145]: Accepted publickey for ubuntu from 13.48.4.203 port 8276 ssh2:
May 20 06:01:31 ip-172-31-32-12 sshd-session[1145]: pam_unix(sshd:session): session opened for user ubuntu(uid=100
ubuntu@ip-172-31-32-12:~$
```

Command 6 — View System Logs

tail -n 50 /var/log/syslog

```
ubuntu@ip-172-31-32-12:~$ tail -n 50 /var/log/syslog
2026-05-20T06:01:31.419261+00:00 ip-172-31-32-12 systemd[1448]: Listening on gpg-agent-browser.socket - GnuPG crypt
.
2026-05-20T06:01:31.419367+00:00 ip-172-31-32-12 systemd[1448]: Listening on gpg-agent-extra.socket - GnuPG cryptog
2026-05-20T06:01:31.419612+00:00 ip-172-31-32-12 systemd[1448]: Starting gpg-agent-ssh.socket - GnuPG cryptographic
2026-05-20T06:01:31.423581+00:00 ip-172-31-32-12 systemd[1448]: Starting gpg-agent.socket - GnuPG cryptographic age
2026-05-20T06:01:31.423871+00:00 ip-172-31-32-12 systemd[1448]: Listening on keyboxd.socket - GnuPG public key mana
2026-05-20T06:01:31.424184+00:00 ip-172-31-32-12 systemd[1448]: Listening on pk-debconf-helper.socket - debconf com
2026-05-20T06:01:31.424435+00:00 ip-172-31-32-12 systemd[1448]: Listening on snapd.session-agent.socket - REST API
2026-05-20T06:01:31.427419+00:00 ip-172-31-32-12 systemd[1448]: Starting ssh-agent.socket - OpenSSH Agent socket...
2026-05-20T06:01:31.432008+00:00 ip-172-31-32-12 systemd[1448]: Listening on systemd-ask-password.socket - Query th
2026-05-20T06:01:31.439868+00:00 ip-172-31-32-12 systemd[1448]: Listening on dbus.socket - D-Bus User Message Bus S
2026-05-20T06:01:31.451557+00:00 ip-172-31-32-12 systemd[1448]: Listening on gpg-agent.socket - GnuPG cryptographic
2026-05-20T06:01:31.452776+00:00 ip-172-31-32-12 systemd[1448]: Listening on ssh-agent.socket - OpenSSH Agent socke
2026-05-20T06:01:31.458817+00:00 ip-172-31-32-12 systemd[1448]: Listening on gpg-agent-ssh.socket - GnuPG cryptogra
2026-05-20T06:01:31.458913+00:00 ip-172-31-32-12 systemd[1448]: Reached target sockets.target - Sockets.
2026-05-20T06:01:31.459007+00:00 ip-172-31-32-12 systemd[1448]: Reached target basic.target - Basic System.
2026-05-20T06:01:31.459097+00:00 ip-172-31-32-12 systemd[1448]: Reached target default.target - Main User Target.
2026-05-20T06:01:31.459226+00:00 ip-172-31-32-12 systemd[1448]: Startup finished in 176ms.
2026-05-20T06:01:31.459364+00:00 ip-172-31-32-12 systemd[1]: Started user@1000.service - User Manager for UID 1000.
```

6. Mini Troubleshooting Section

Add:

```
□## Mini Troubleshooting Steps
```

```
□
```

Example:

```
□### Problem
```

```
SSH service was not running.
```

```
### Troubleshooting Steps
```

```
Checked service status:
```

```
```bash
systemctl status ssh
```
```

```
Started the service:
```

```
```bash
sudo systemctl start ssh
```
```

```
Verified again:
```

```
```bash
systemctl status ssh
```
```

```
### Result
```

```
Service became active and running.
```

```
□
```

```
lines 1-25/25 (END)
```

- `journalctl -u ssh --no-pager | tail -n 20`
- `tail -n 50 /var/log/syslog`